

## Compound angle equations



1 Consider the equation  $\cos(\theta + 20^\circ) = \frac{1}{\sqrt{2}}$  for  $0^\circ \leq \theta \leq 360^\circ$ .

a If  $\alpha = \theta + 20^\circ$ , find the solutions of  $\cos \alpha = \frac{1}{\sqrt{2}}$  for  $20^\circ \leq \alpha \leq 380^\circ$ .

b Hence, solve  $\cos(\theta + 20^\circ) = \frac{1}{\sqrt{2}}$ .

2 Consider the equation  $\cos(\theta - 100^\circ) = \frac{1}{2}$  for  $0^\circ \leq \theta \leq 360^\circ$ .

a If  $\alpha = \theta - 100^\circ$ , find the solutions of  $\cos \alpha = \frac{1}{2}$  for  $-100^\circ \leq \alpha \leq 260^\circ$ .

b Hence, solve  $\cos(\theta - 100^\circ) = \frac{1}{2}$ .

3 Consider the equation  $\sin(\theta + 35^\circ) = \frac{1}{2}$  for  $0^\circ \leq \theta \leq 360^\circ$ .

a If  $\alpha = \theta + 35^\circ$ , find the solutions of  $\sin \alpha = \frac{1}{2}$  for  $35^\circ \leq \alpha \leq 395^\circ$ .

b Hence, solve  $\sin(\theta + 35^\circ) = \frac{1}{2}$ .

4 Consider the equation  $\sin(\theta + 25^\circ) = \frac{\sqrt{3}}{2}$  for  $0^\circ \leq \theta \leq 360^\circ$ .

a If  $\alpha = \theta + 25^\circ$ , find the solutions of  $\sin \alpha = \frac{\sqrt{3}}{2}$  for  $25^\circ \leq \alpha \leq 385^\circ$ .

b Hence, solve  $\sin(\theta + 25^\circ) = \frac{\sqrt{3}}{2}$ .

5 Consider the equation  $\tan(\theta + 15^\circ) = \frac{1}{\sqrt{3}}$  for  $0^\circ \leq \theta \leq 360^\circ$ .

a If  $\alpha = \theta + 15^\circ$ , find the solutions of  $\tan \alpha = \frac{1}{\sqrt{3}}$  for  $15^\circ \leq \alpha \leq 375^\circ$ .

b Hence, solve  $\tan(\theta + 15^\circ) = \frac{1}{\sqrt{3}}$ .

6 Solve the following equations for  $0^\circ \leq \theta \leq 360^\circ$ :

a  $\cos(\theta + 60^\circ) = 1$

b  $\cos(\theta - 75^\circ) = -\frac{1}{\sqrt{2}}$

c  $\sin(\theta + 45^\circ) = -\frac{\sqrt{3}}{2}$

d  $\tan(\theta - 45^\circ) = \frac{1}{\sqrt{3}}$

e  $\cos(\theta - 90^\circ) = 1$

f  $2 \sin(\theta - 60^\circ) = -\sqrt{3}$



**7** Solve the following equations for  $0^\circ \leq \theta \leq 360^\circ$ . Round your answers to two decimal places if necessary.

**a**  $\cos 3\theta = \frac{1}{\sqrt{2}}$

**b**  $\sin 2\theta = -\frac{1}{2}$

**c**  $\cos 2\theta = 0.7$

**d**  $2 \sin 3\theta - \sqrt{2} = 0$

**e**  $\sin\left(\frac{\theta}{2}\right) = \frac{\sqrt{3}}{2}$

**f**  $\tan 3\theta = -1$

**8** Solve the following equations for  $0^\circ \leq x \leq 180^\circ$ :

**a**  $\tan 4x = \sqrt{3}$

**b**  $\sin 2x = \frac{\sqrt{3}}{2}$

**c**  $\cos 3x = -\frac{1}{\sqrt{2}}$

**d**  $\tan\left(\frac{x}{3}\right) = 1$

**9** Solve the following equations for  $-180^\circ \leq x \leq 180^\circ$ :

**a**  $\cos 3x = -\frac{1}{\sqrt{2}}$

**b**  $\sin 2x = \frac{1}{2}$

**c**  $\tan\left(\frac{x}{2}\right) = \pm\sqrt{3}$

**d**  $\sin 3x - 1 = 0$

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## Squared equations

**10** Solve the following equations for  $0^\circ \leq \theta \leq 360^\circ$ . Round your answers to two decimal places if necessary.

**a**  $2 \sin^2 \theta = 1$

**b**  $\sin^2 \theta = \frac{3}{4}$

**c**  $\cos^2 \theta = \frac{1}{2}$

**d**  $\tan^2 \theta = \frac{1}{3}$

**e**  $4 \sin^2 \theta = 1$

**f**  $\cos^2\left(\frac{\theta}{2}\right) - 1 = 0$

**g**  $\sin^2 2\theta = \frac{1}{2}$

**h**  $\cos^2 3\theta = \frac{3}{4}$

**11** Solve the following equations for the given domain:

**a**  $\cos^2 \theta = \frac{3}{4}$  for  $0^\circ < \theta < 90^\circ$

**b**  $4 \sin^2 \theta = 3$  for  $90^\circ \leq \theta \leq 270^\circ$



## Factorisation

- 12** State the number of solutions for  $\theta$  of the equation  $\left(\sin \theta + \frac{\sqrt{3}}{2}\right) \left(\cos \theta + \frac{\sqrt{3}}{2}\right) = 0$  for  $0^\circ < \theta < 90^\circ$ .
- 13** Solve the following equations for  $0^\circ \leq \theta \leq 360^\circ$ . Round your answer to one decimal place if necessary.
- a**  $\tan^2(\theta) - 2 \tan \theta + 1 = 0$                       **b**  $4 \cos^3(\theta) = 3 \cos \theta$
- c**  $5 \sin^2(\theta) + 8 \sin \theta \cos \theta - 4 \cos^2(\theta) = 0$                       **d**  $7 \cos^2(\theta) + 3 \cos \theta = 3$
- e**  $9 \sin^2(x) + 5 \sin x = 3$                       **f**  $\cos \theta \tan \theta = \cos \theta$
- g**  $8 \tan^2(\theta) \cos \theta - 4 \tan^2(\theta) = 0$
- 14** Solve the following equations for  $0^\circ < \theta < 90^\circ$ :
- a**  $\left(\sin \theta - \frac{\sqrt{3}}{2}\right) \left(\cos \theta - \frac{1}{\sqrt{2}}\right) = 0$                       **b**  $\left(\sin \theta + \frac{1}{\sqrt{2}}\right) \left(\tan \theta - \frac{1}{\sqrt{3}}\right) = 0$
- 15** Solve the following equations for the given domain:
- a**  $\left(\tan \theta - \frac{1}{\sqrt{3}}\right) \left(\cos \theta + \frac{1}{\sqrt{2}}\right) = 0$  for  $180^\circ \leq \theta \leq 270^\circ$
- b**  $\left(\sin \theta - \frac{1}{2}\right) \left(\cos \theta - \frac{1}{\sqrt{2}}\right) = 0$  for  $270^\circ \leq \theta \leq 360^\circ$
- c**  $\tan^2(\theta) = \sqrt{3} \tan \theta$  for  $-90^\circ < \theta < 90^\circ$
- d**  $2 \sin^2(\theta) - \sin \theta = 0$  for  $-180^\circ \leq \theta \leq 180^\circ$
- 16** Deborah is solving the equation  $2 \sin^2 \theta + 7 \sin \theta + 5 = 0$ . After some factorisation, she arrives at the pair of equations  $\sin \theta + 1 = 0$  and  $2 \sin \theta + 5 = 0$ . Which of the two equations has a solution?



## Equations and trigonometric identities

**17** Solve the equation  $3 \sin \theta = 3\sqrt{3} \cos \theta$  for  $0 < \theta < 90$ .

**18** Solve the following equations for  $0^\circ \leq \theta < 360^\circ$ . Round your answers to the nearest degree if necessary.

**a**  $2 \cos^2 (\theta) = 2 - \sin \theta$

**b**  $\sin^5 (\theta) \cos^3 (\theta) = 0$

**c**  $\cos^2 (\theta) - 8 \sin \theta \cos \theta + 3 = 0$

**d**  $\frac{1 - \tan^2 (\theta)}{1 + \tan^2 (\theta)} + \cos \theta = 0$